



Data Link Layer : MAC sublayer

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Multiple Access Problem

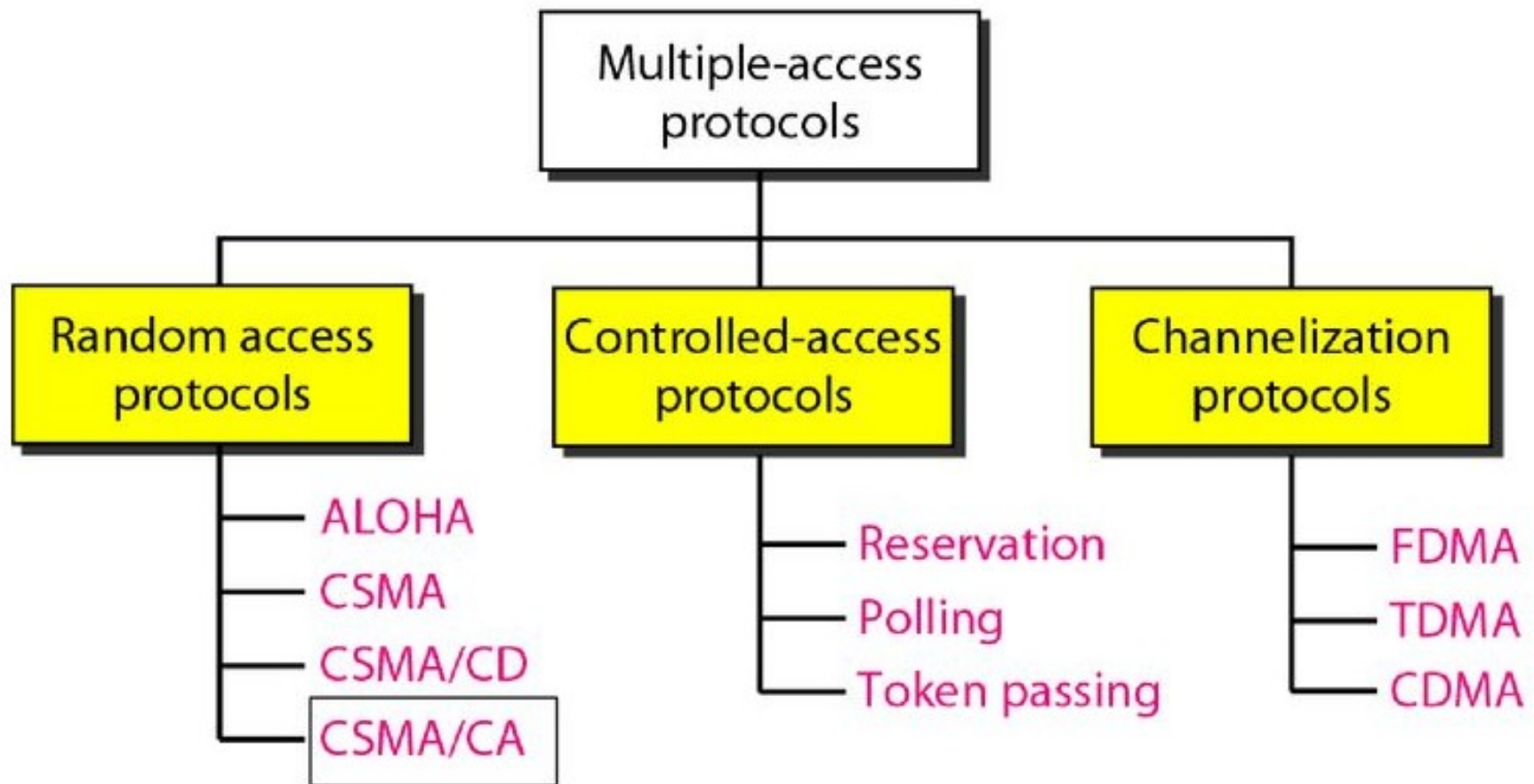
- Some real-world scenarios that can be visualized as multiple access problem
 - Question hours in assembly
 - Group Discussion
 - Lavatory in Airplane



Multiple Access in Communication

- Two or more nodes can transmit frames at the same time
- When transmission frames collides at all receivers, it does not make any sense
- Frames are lost and bandwidth is waster
- Thus, When stations share a common link (also called multipoint or broadcast link), a **multiple-access protocol** is required to coordinate the access to the link

Multiple Access protocols





Random Access

- Also called **contention methods**
 - Stations compete with one another to access the medium
 - No station is superior to another station and none is assigned the control over another
 - No station permits, or does not permit, another station to send
- When a station that has data to send it uses a procedure defined by the protocol to decide, whether or not to send
- Depends on the state of the medium (idle or busy)
- If more than one station tries to send, there is an access conflict-collision and the frames will be either destroyed or modified



ALOHA

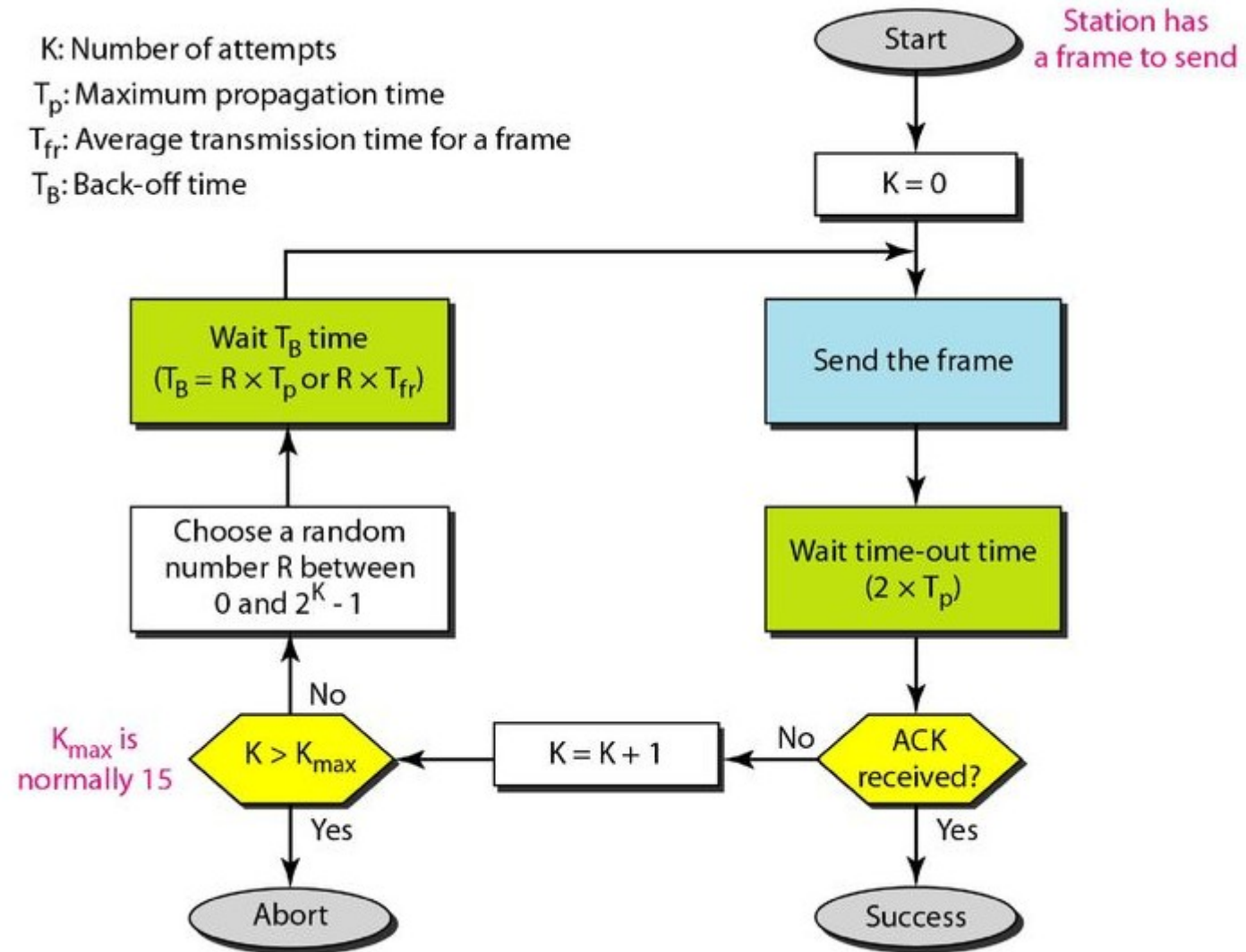
- The earliest random access method, was developed at the University of Hawaii in early 1970
- Designed for a radio (wireless) LAN, but it can be used on any shared medium
- The original version is called **pure ALOHA**



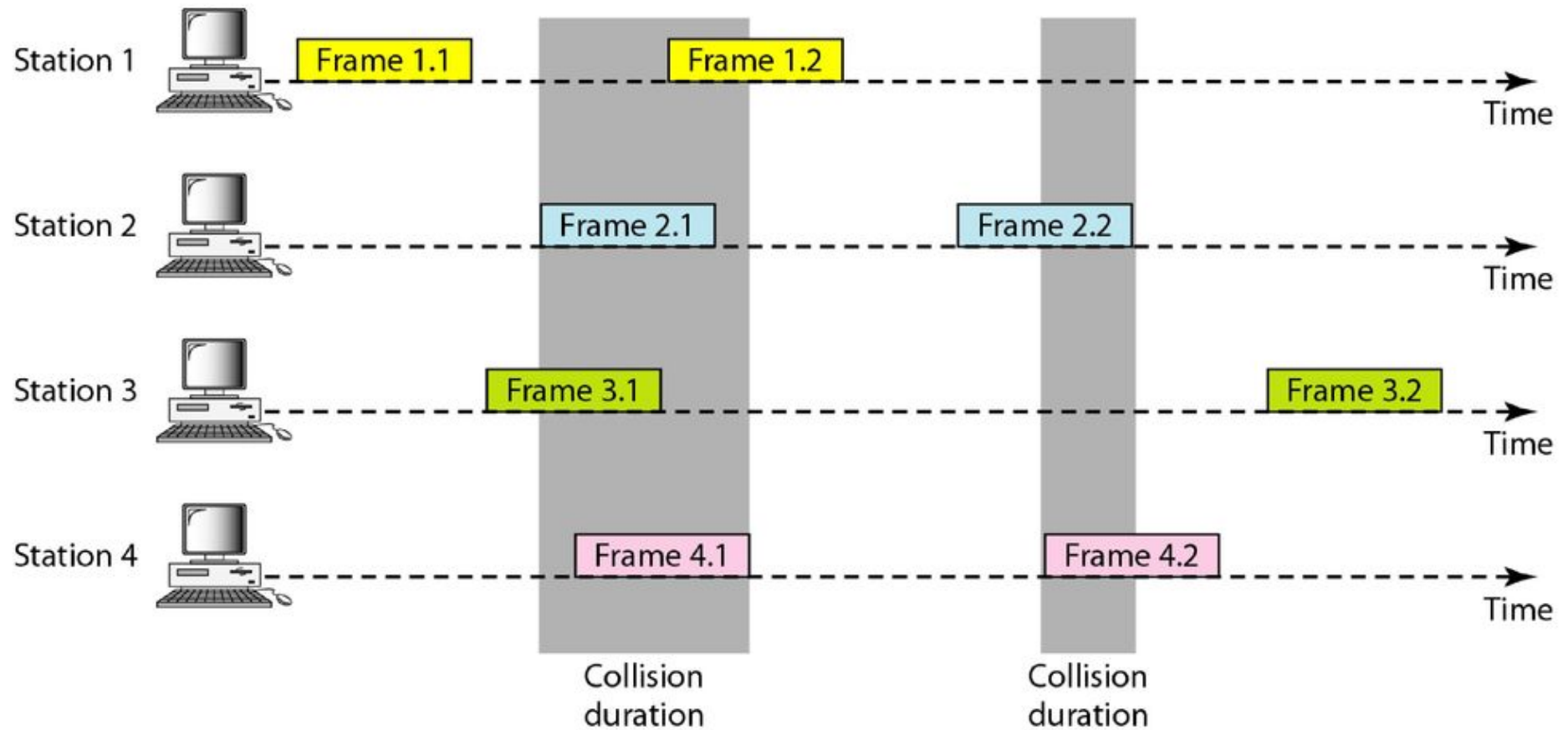
Pure ALOHA

- When a frame arrives, then it is **immediately sent** to the broadcast channel at full rate
- if(collision detected)
 - **wait**(random period of time);
 - **retransmitting** the frame;
- Repeat the retransmission process until frame is transmitted successfully without collision
- Sender expect an ACK from receiver within a finite waiting time, otherwise retransmission is initiated

Procedure in pure ALOHA



Collision in Pure ALOHA



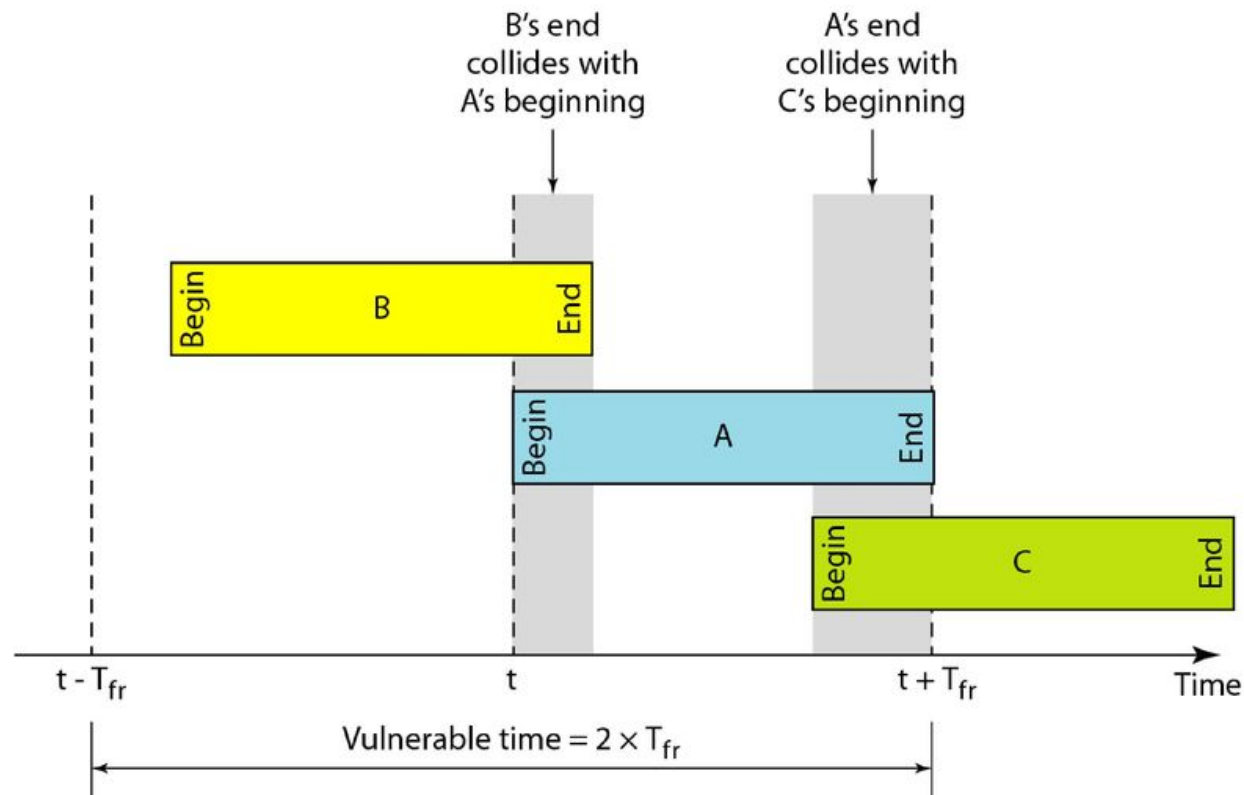


Binary exponential back-off

- The time-out period is equal to the maximum possible round-trip propagation delay (twice the maximum transmission time)
- The **back-off time T_B** is a random value that normally depends on K (the number of attempted unsuccessful transmissions)
- Binary exponential back-off is a simplest way to determine T_B
 - a multiplier in the range 0 to $2^k - 1$ is randomly chosen
 - It is multiplied by T_p (maximum propagation time) or T_{fr} (the average time required to send out a frame) to obtain T_B
- The range of the random numbers increases after each collision until K reached K_{max}
- The value of K_{max} is usually chosen as 15

Vulnerable time of pure ALOHA

- The vulnerable time is the duration in which there is a possibility of collision
- Pure ALOHA vulnerable time = $2 \times T_{fr}$





Throughput of pure ALOHA

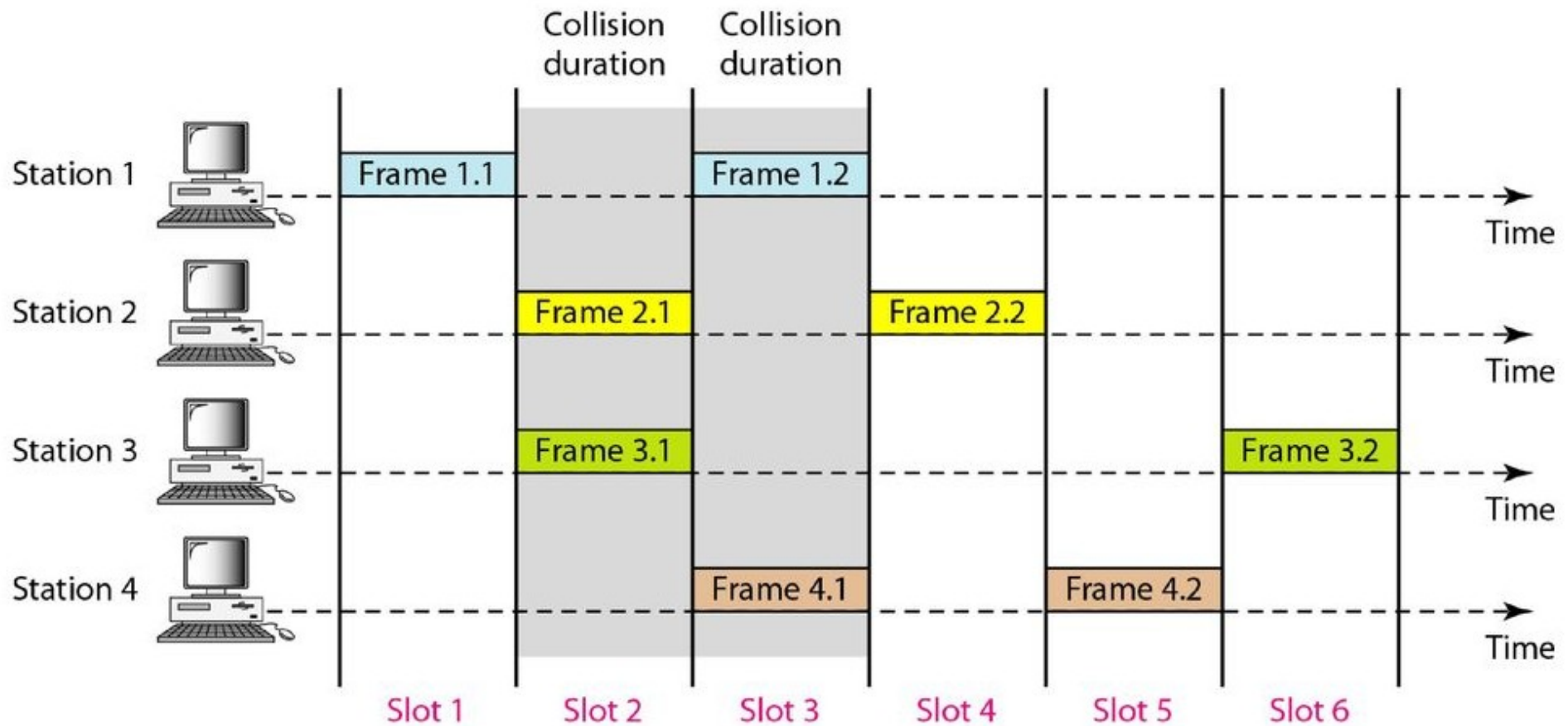
- Let G the average number of frames generated by the system during one frame transmission time, then it can be proved that the average number of successful transmissions for pure ALOHA is $S = G \times e^{-2G}$.



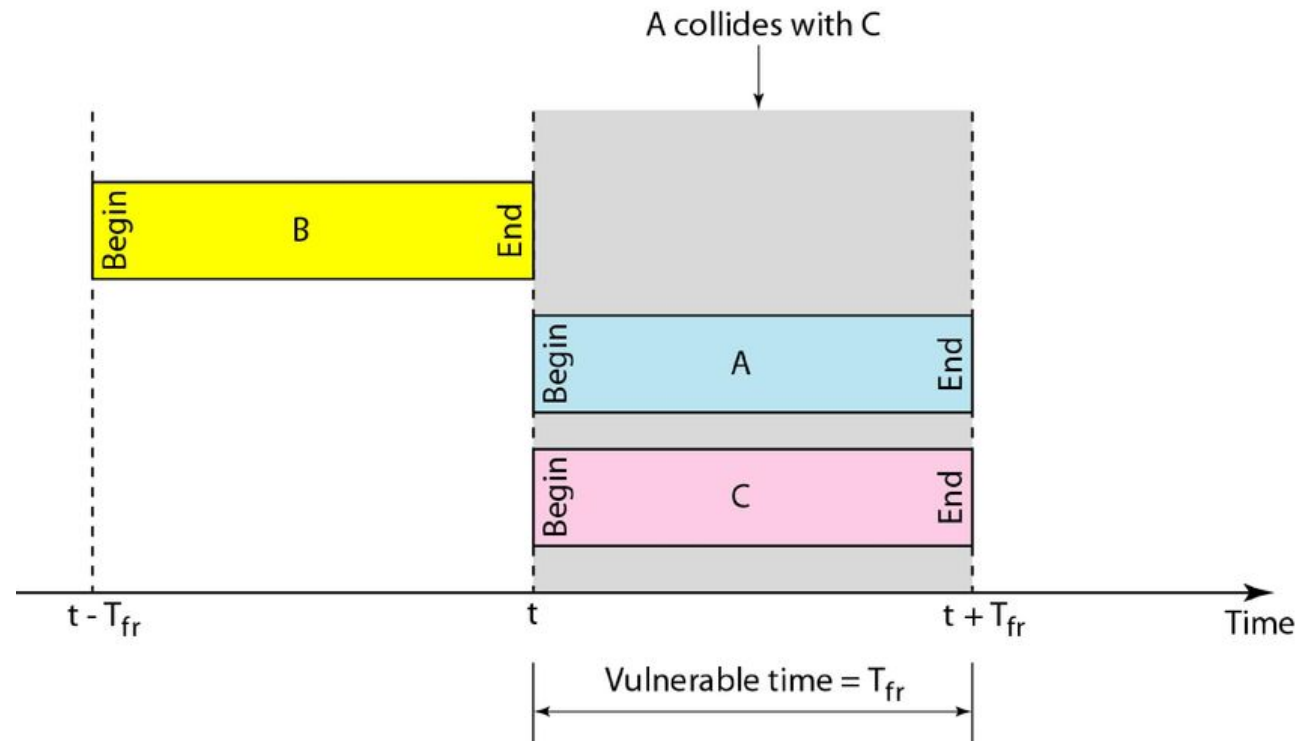
Slotted ALOHA

- Pure ALOHA has a vulnerable time of $2 \times T_{fr}$ **because** “there is no rule that defines when the station can send”
- In slotted ALOHA the time is divided into slots of T_{fr} and force the station to send only at the **beginning** of the time slot
- If collision occurs then the station waits for random waiting time; when timer expires it is ready to send frame at the beginning of the **next time slot**
- Student task - (Update the procedure of pure ALOHA in slide 8 to define the procedure of slotted ALOHA)

Collision in slotted ALOHA



Vulnerable time of slotted ALOHA



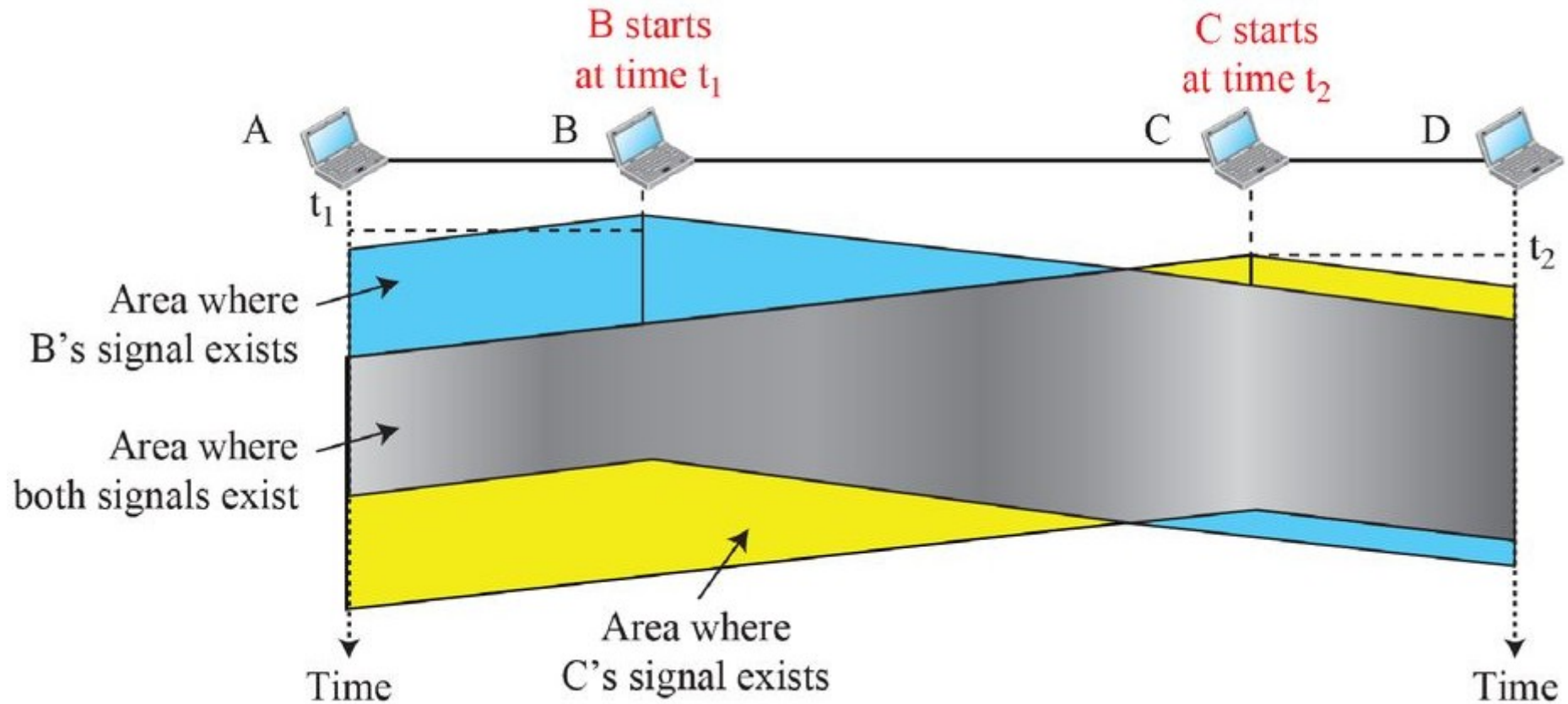
- Slotted ALOHA vulnerable time = T_{fr}
- Throughput - the average number of successful transmissions for slotted ALOHA is $S = G \times e^{-G}$



Carrier Sense Multiple Access (CSMA)

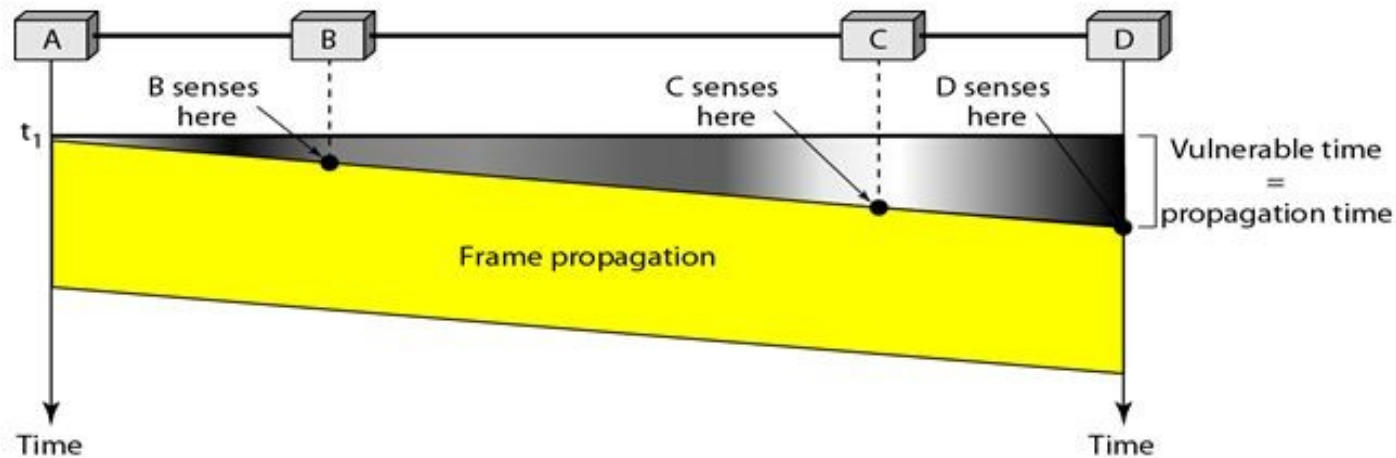
- The chance of collision can be reduced if a station senses the medium before trying to use it
- Carrier sense multiple access (CSMA) requires that each station first **listen to the medium** (or check the state of the medium) before sending
- CSMA can reduce the possibility of collision, but it cannot eliminate it
- The possibility of **collision** still exists because of **propagation delay**
 - When a station sends a frame, it still takes time (although very short) for the first bit to reach every station and for every station to sense it

Space/time model of the collision in CSMA



Vulnerable time of CSMA

- The vulnerable time for CSMA is the propagation time T_p
 - This is the time needed for a signal to propagate from one end of the medium to the other

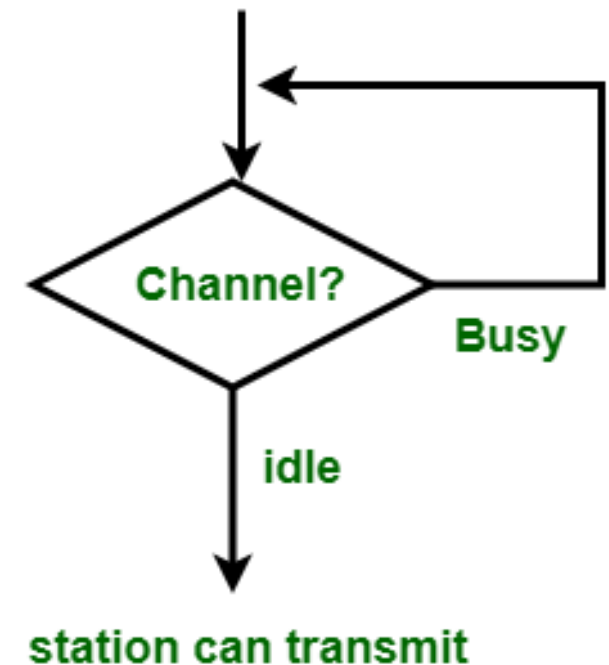
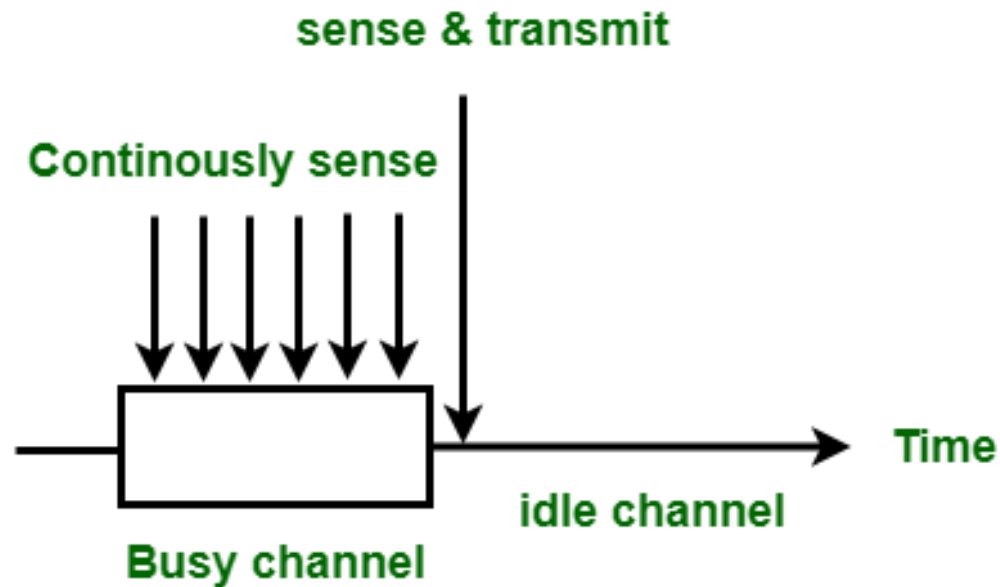




Persistent CSMA

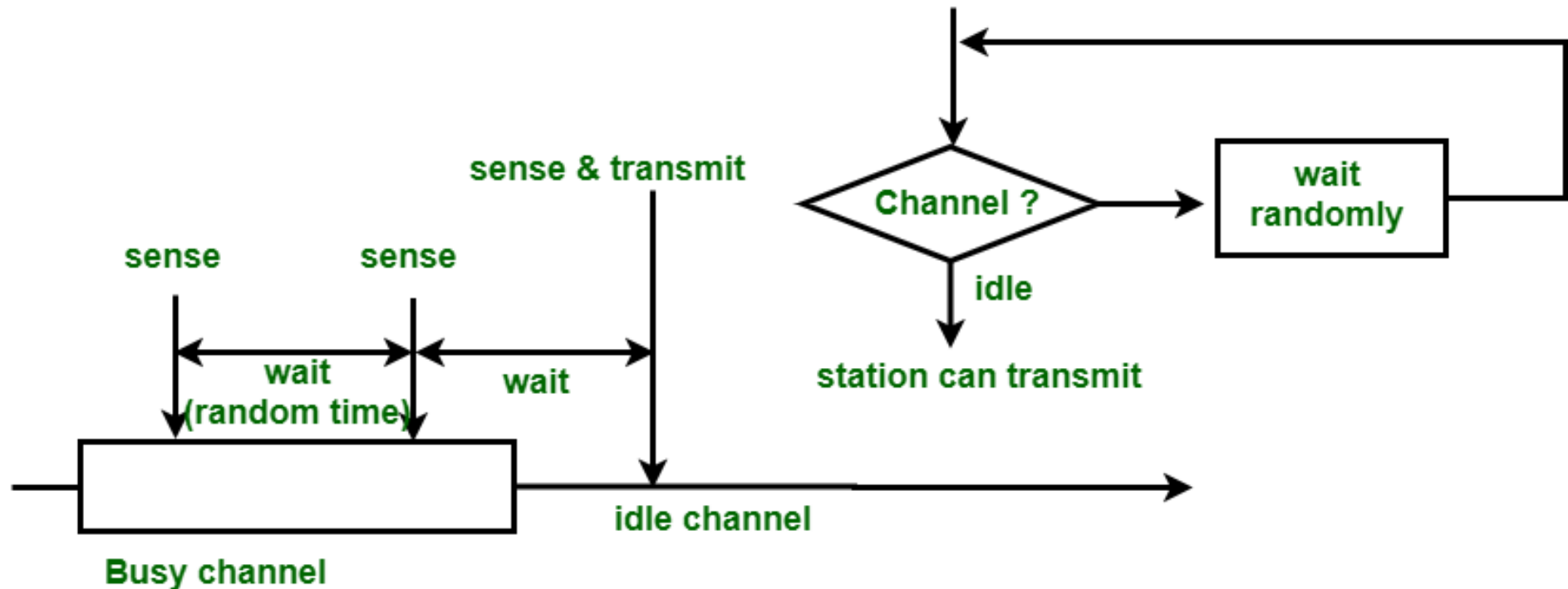
- How the station must act when channel is idle?
- Three methods
 - 1-persistent method
 - Nonpersistent method
 - The p-persistent method

1-persistent CSMA



- Continuously sense the channel
- When channel is idle, sends its frame immediately (probability of 1)

non-persistent CSMA



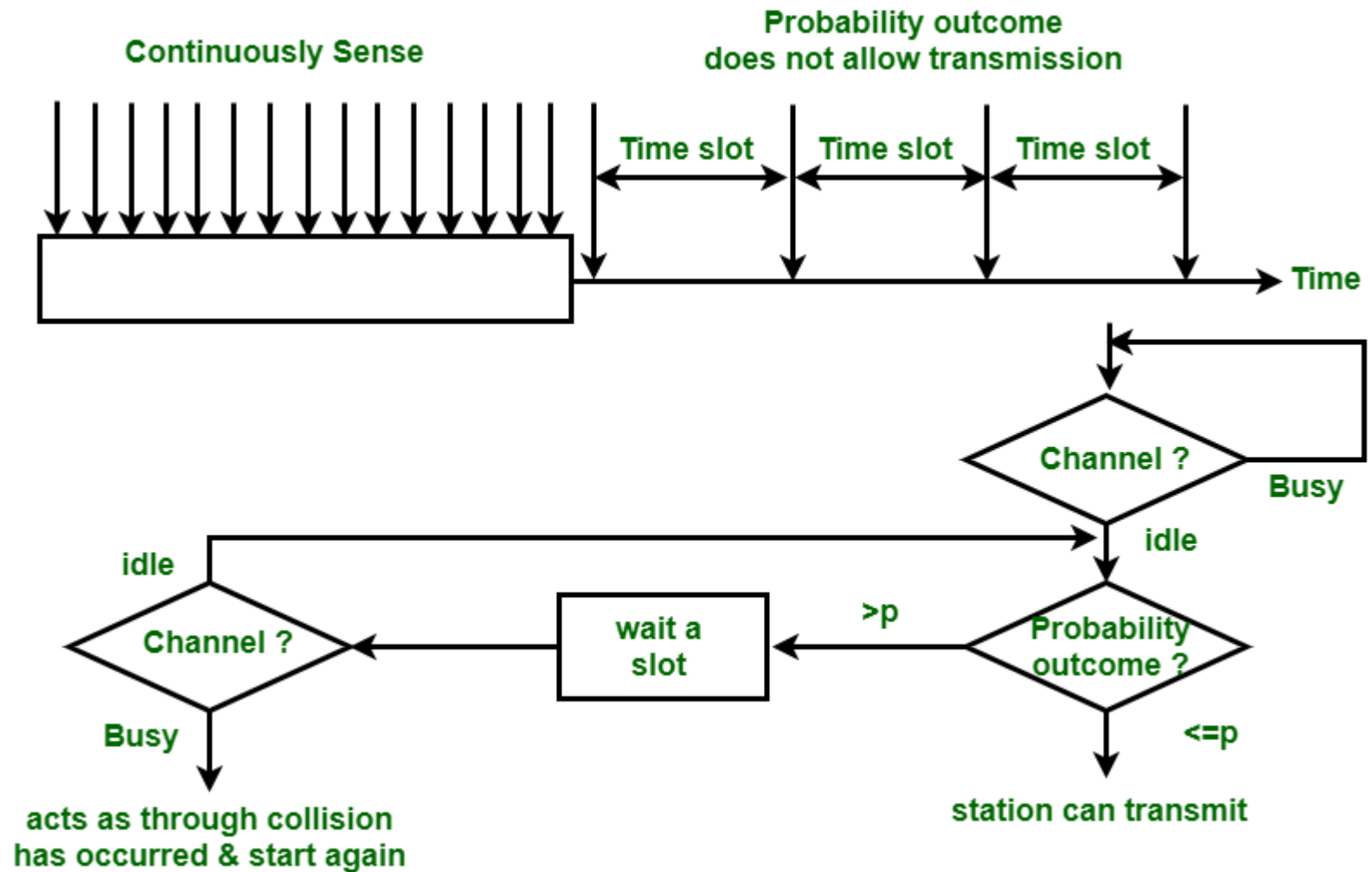
- When channel is idle, sends its frame immediately. If the line is not idle, it waits a random amount of time and then senses the line again



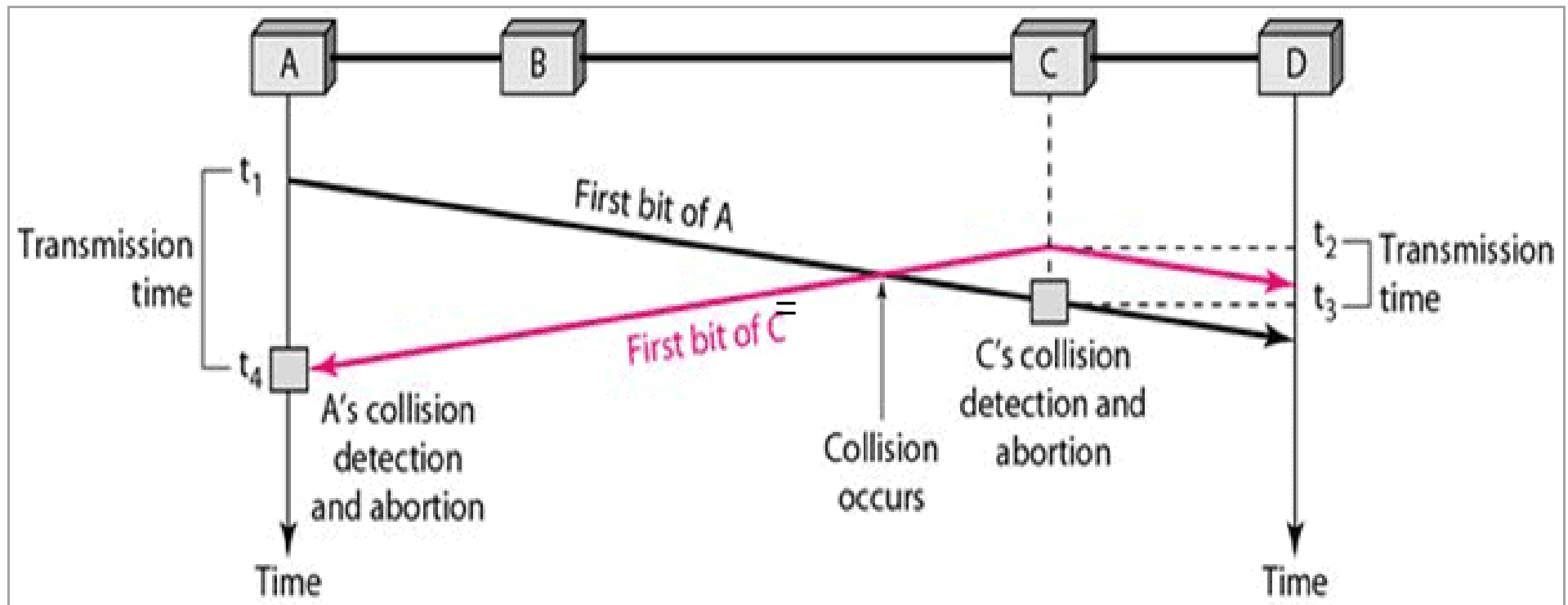
p-persistent CSMA

- It is used if the channel has time slots with a slot duration equal to or greater than the maximum propagation time
- Combines the advantages of the other two strategies
- It reduces the chance of collision and improves efficiency
- If the channel is idle, then
 - With probability p , the station sends its frame.
 - With probability $q = 1 - p$, the station waits for the beginning of the next time slot and checks the line again.
 - If the line is idle, it goes to step 1.
 - If the line is busy, it acts as though a collision has occurred and uses the back-off procedure

p-persistent CSMA



Carrier Sense Multiple Access with Collision Detection (CSMA/CD)



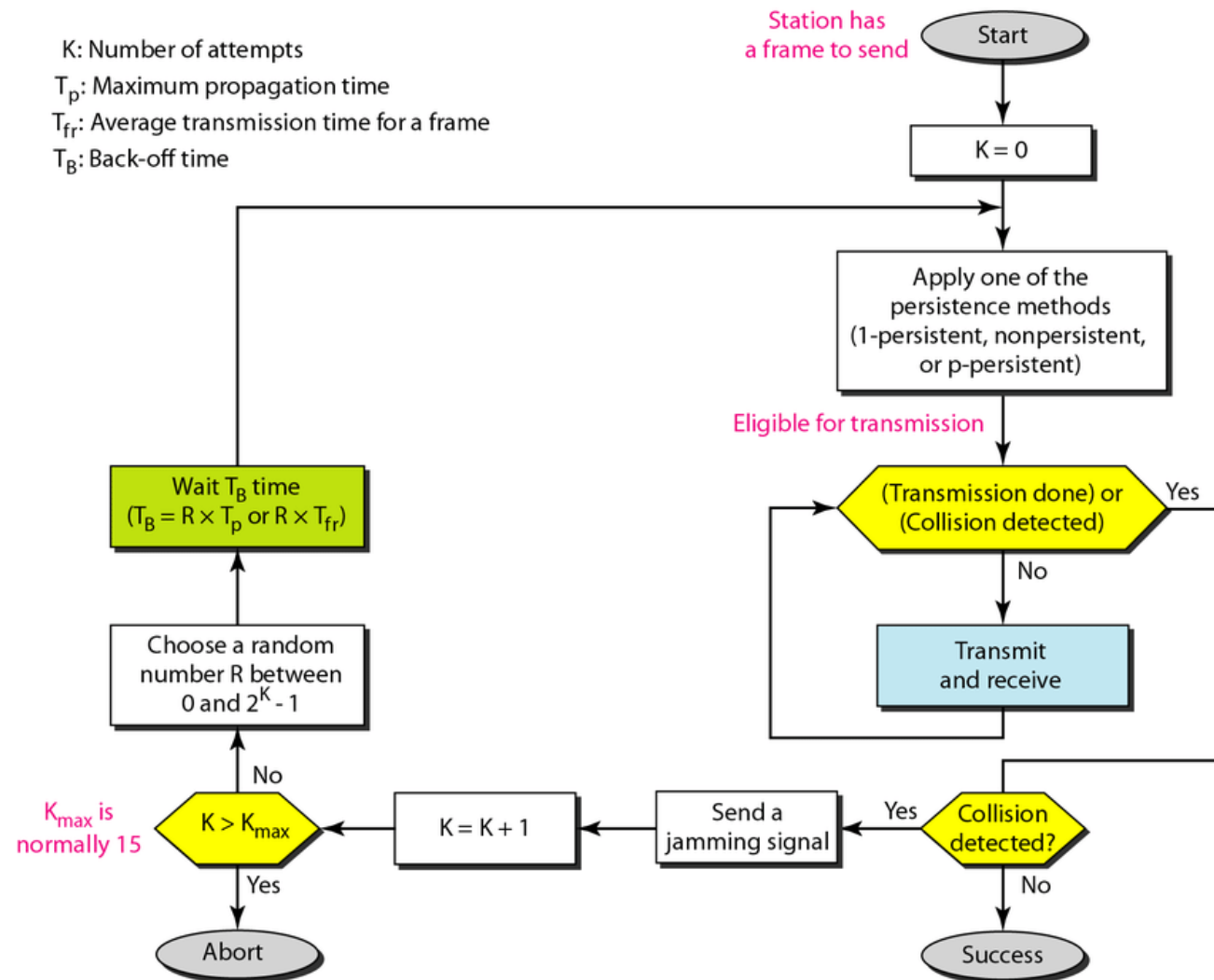
CSMA/CD augments the algorithm to handle the collision

Carrier Sense Multiple Access with Collision Detection (CSMA/CD)



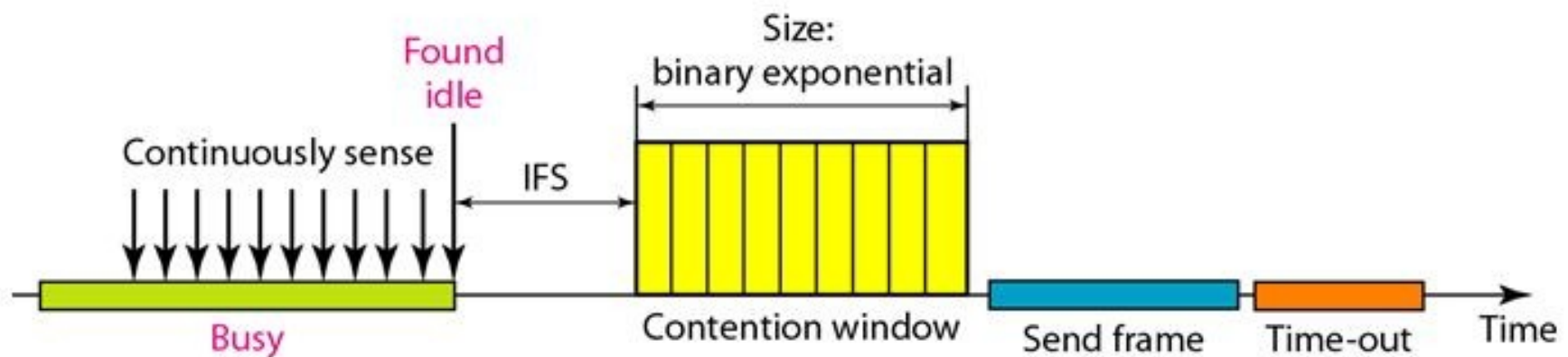
- Minimum Frame Size
 - The frame transmission time T_{fr} must be at least two times the maximum propagation time T_p
 - The signal from the first takes time T_p to reach the second, and the effect of the collision takes another time T_p to reach the first
- Procedure
 - Need a persistence process to sense the channel
 - Do not send the entire frame and then look for a collision
 - The station transmits and receives **continuously and simultaneously** (using two different ports)
 - Sending of a **short jamming signal** that enforces the collision in case other stations have not yet sensed the collision

CSMA/CD

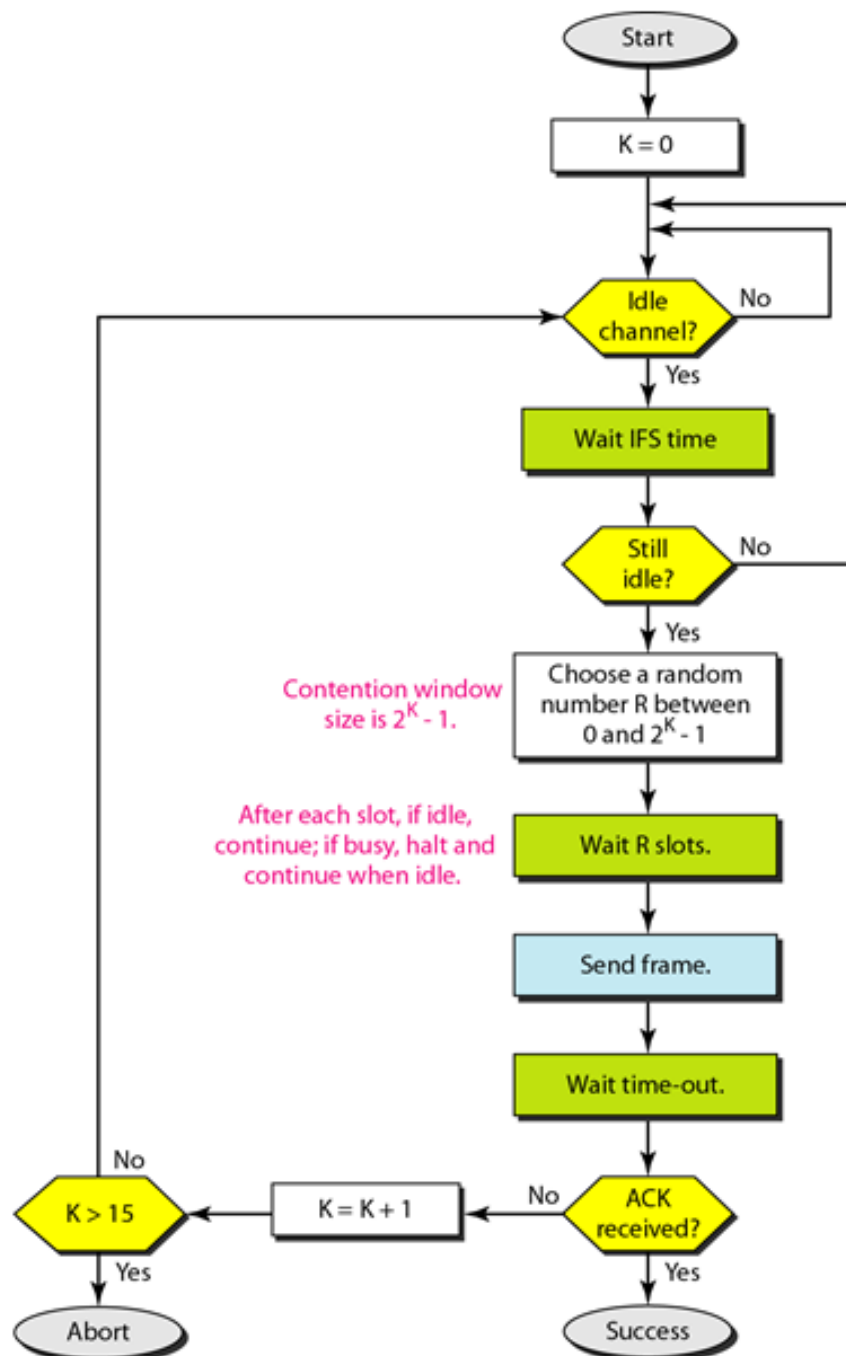


Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)

- Collision detection is not useful in wireless channel
- Collisions are **avoided** through the use of CSMA/CA's three strategies:
 - The interframe space
 - The contention window, and
 - Acknowledgments



CSMA/CA



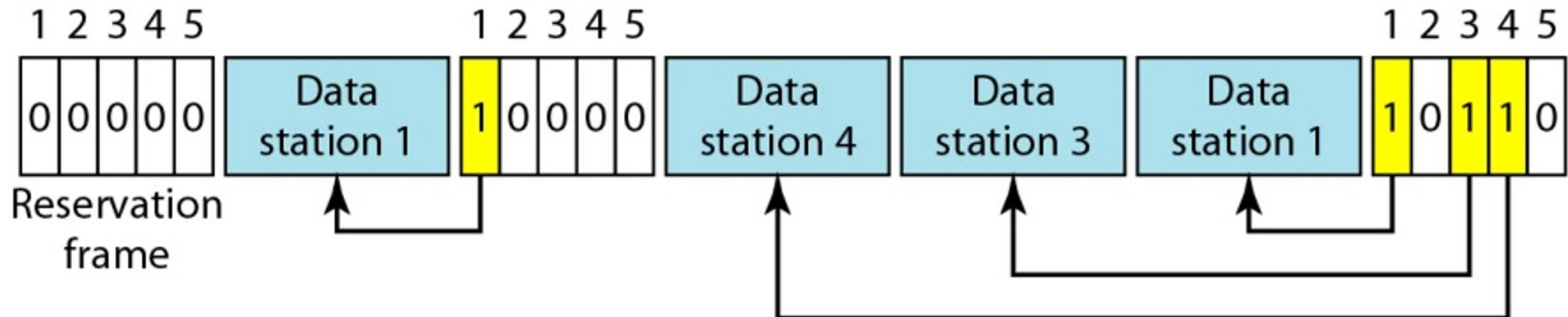


Controlled Access

- The stations consult one another to find which station has the right to send. A station cannot send unless it has been authorized by other stations
 - Reservation
 - Polling
 - Token Passing

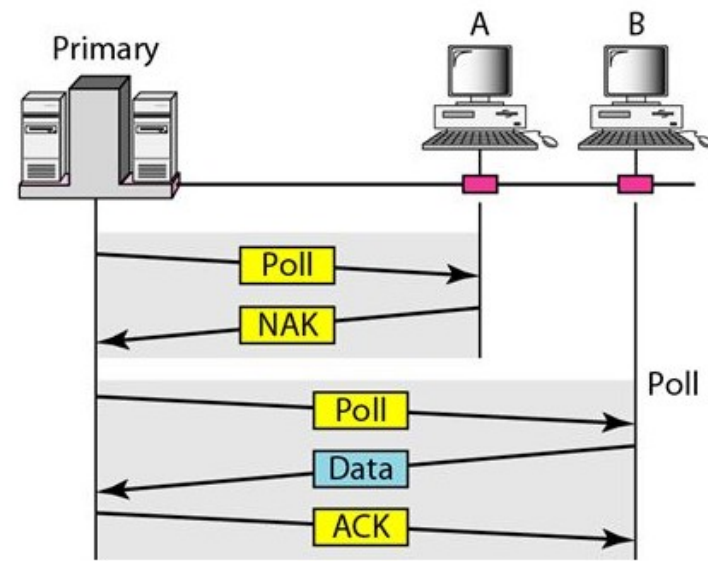
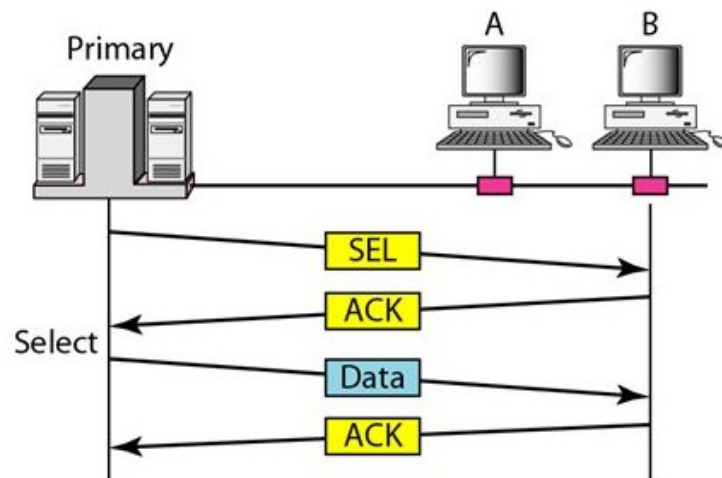
Reservation

- A station needs to make a reservation before sending data.
- Time is divided into intervals, a reservation frame precedes the data frames sent in that interval.
- If there are N stations in the system, there are exactly N reservation minislots in the reservation frame
- The stations that have made reservations can send their data frames after the reservation frame.



Polling

- One device is designated as a primary station and the other devices are secondary stations
- The primary device controls the link; the secondary devices follow its instructions
- It is up to the primary device to determine which device is allowed to use the channel at a given time
- If the primary wants to receive data, it asks the secondaries if they have anything to send; this is called **poll function**. If the primary wants to send data, it tells the secondary to get ready to receive; this is called **select function**





Token Passing

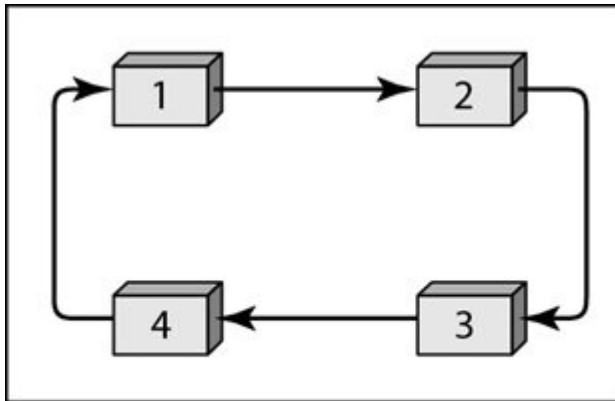
- The stations in a network are organized in a **logical ring**
- The current station is the one that is accessing the channel now
- The right to this access has been passed from the predecessor(previous) to the current station
- The right will be passed to the successor(next) when the current station has no more data to send.
- A special packet called a **token** circulates through the ring. The **possession** of the token gives the station the right to access the channel and send its data



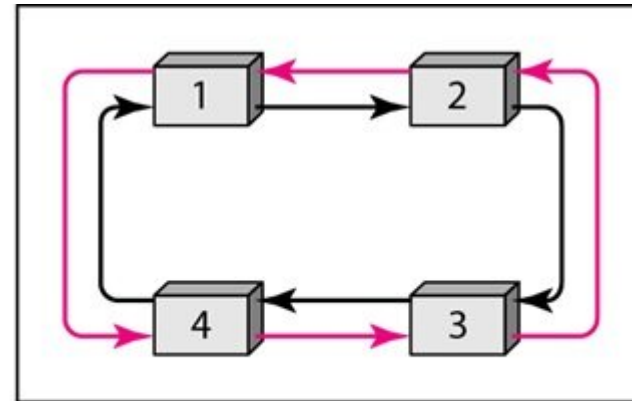
Token Passing

- When a station has some data to send, it waits until it receives the token from its predecessor
- It then holds the token and sends its data. When the station has no more data to send, it releases the token, passing it to the next logical station in the ring
- Token management is needed for the followings:
 - Stations must be limited in the time they can have possession of the token
 - The token must be monitored to ensure it has not been lost or destroyed
 - Make low-priority stations release the token to high-priority stations

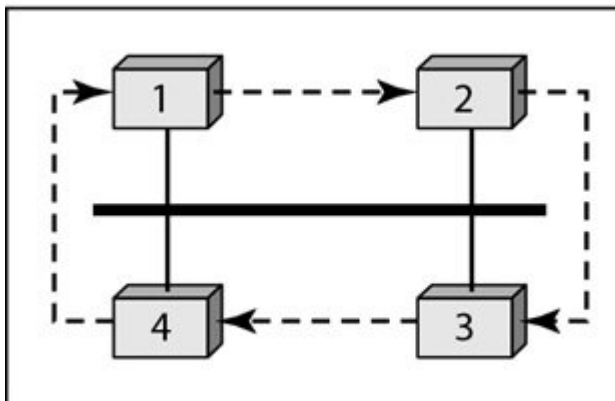
Logical Ring Physical Topology



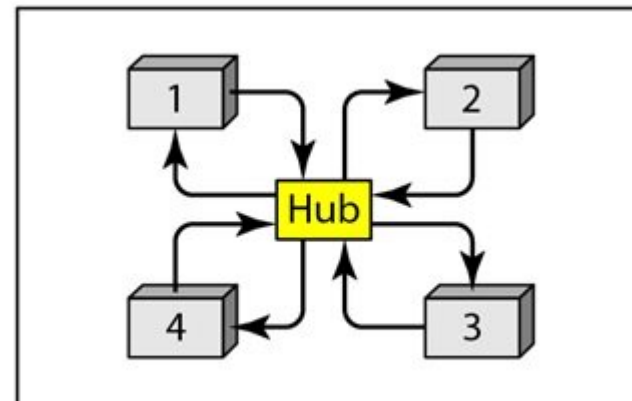
a. Physical ring



b. Dual ring



c. Bus ring



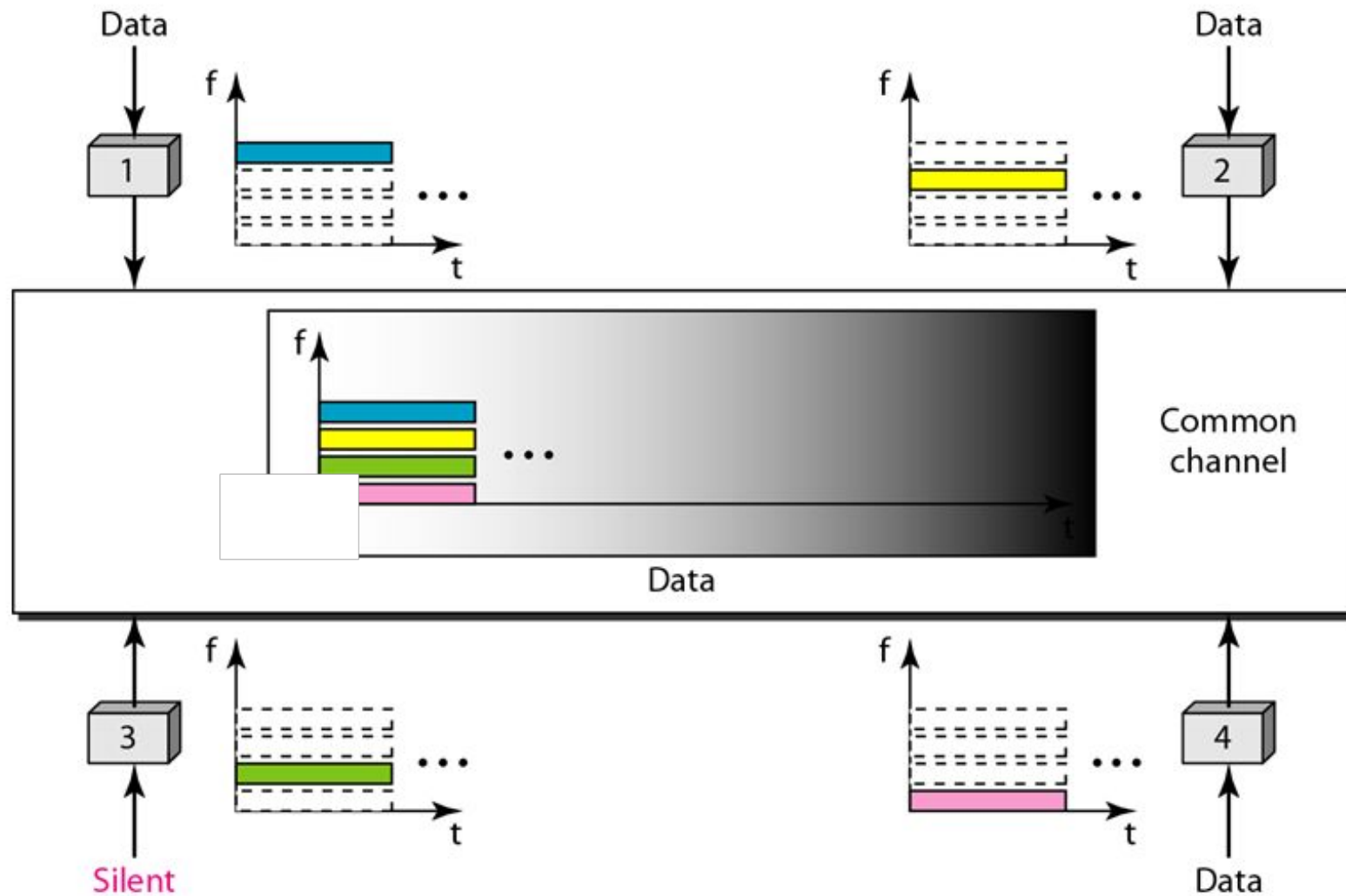
d. Star ring



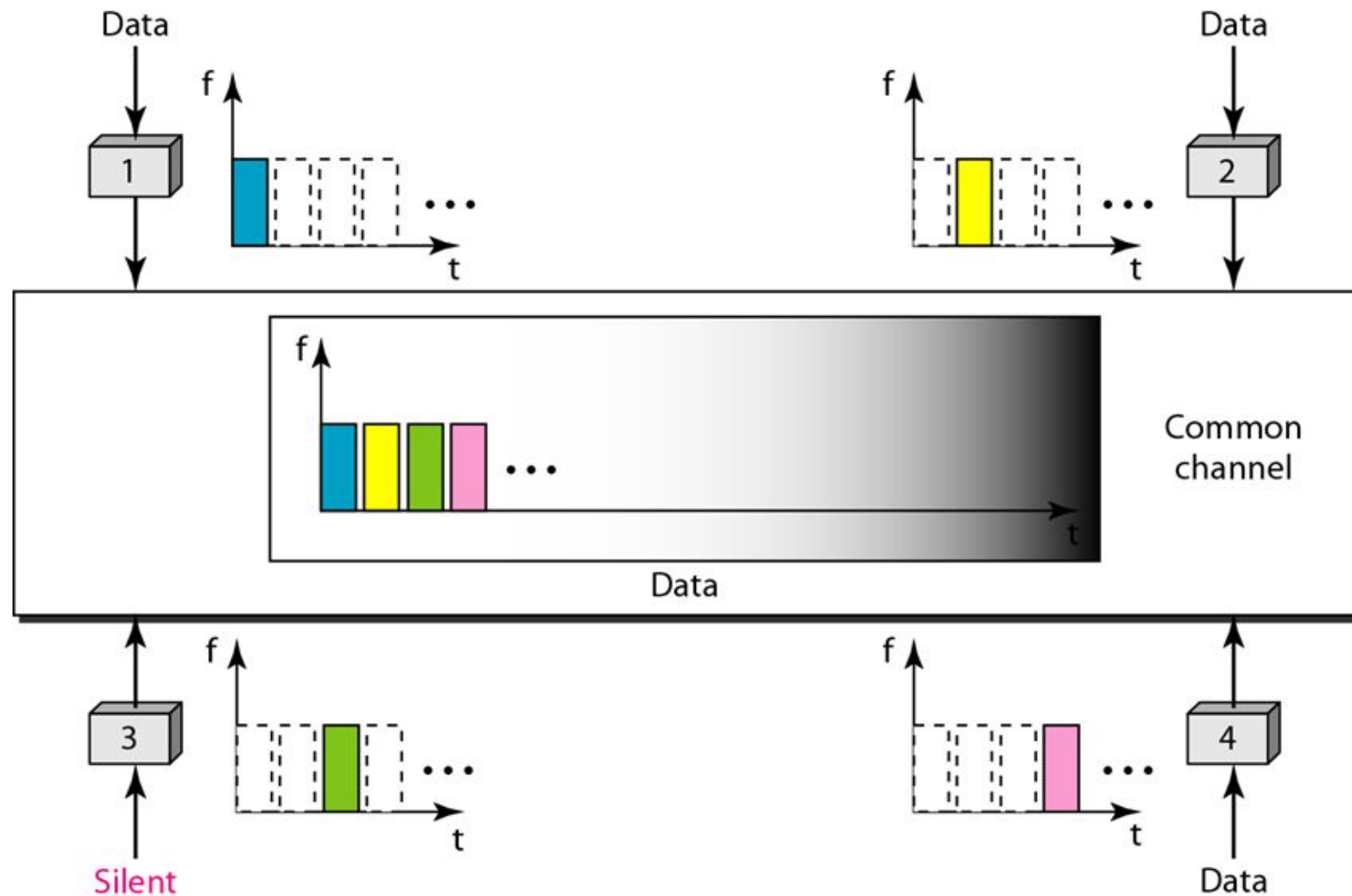
Channelization

- A multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations
 - FDMA
 - TDMA
 - CDMA

Frequency-Division Multiple Access (FDMA)



Time-Division Multiple Access (TDMA)

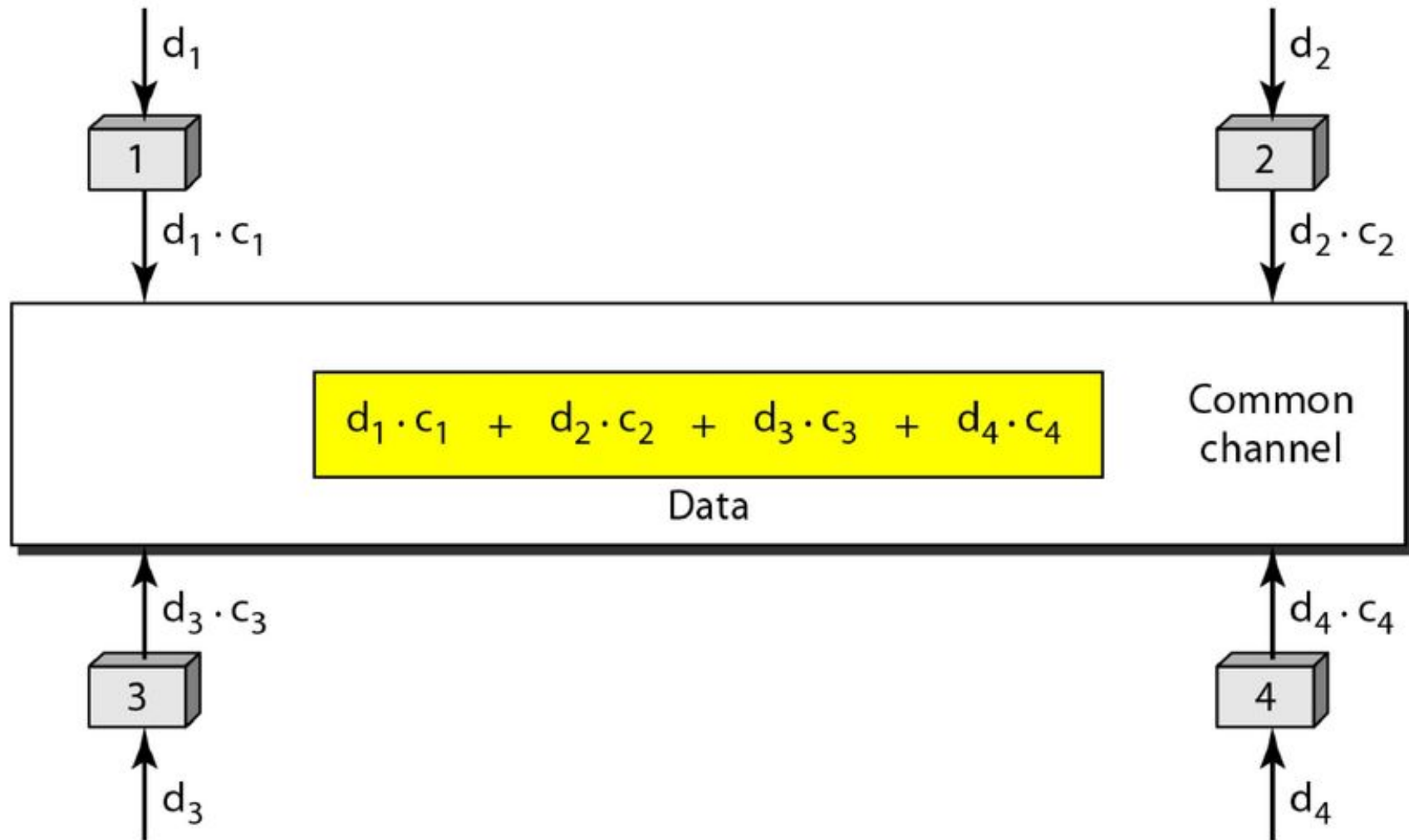




Code-Division Multiple Access (CDMA)

- A spread-spectrum techniques
- A spreading code with higher rate compared to data is generated.
- The data for transmission combined by bitwise XOR with the spreading code
- The data is decoded at the receiver using the same spreading code
- Each station uses different code to modulate their signal

Code-Division Multiple Access (CDMA)



Next : Ethernet